



RIVER VALLEY HIGH SCHOOL

JC 2 PRELIMINARY EXAMINATION

CANDIDATE NAME

CLASS 2 1 J

CENTRE NUMBER S

INDEX NUMBER

H3 CHEMISTRY

9813/03

Paper 1

23 September 2022

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Data Booklet; Insert

READ THESE INSTRUCTIONS FIRST

Write your Centre number, class and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer all questions in the spaces provided on the Question Paper. If additional space is required, you should use the pages at the end of this booklet. The question number must be clearly shown.

Section A

Answer all questions.

Section B

Answer two questions.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use									
Question Number	1	2	3	4	5	6	s.f.	units	Total
Marks	22	18	20	20	20	20			100

This document consists of 36 printed pages.

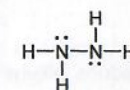
Section A

Answer all questions in this section.

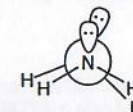
- 1 The information provided in the insert is taken from several published scientific articles. Other published articles may not agree with all of this information.

You should read the whole insert before you start to answer any questions and use the information it contains to answer the questions.

- (a) (i) Calculate a value for x , stating clearly any assumptions made in your calculations. [2]
- (ii) Determine the maximum quantity of energy that can be obtained from 1 kg of ammonia. [1]
- (b) Suggest an advantage of using ammonia over hydrogen as a fuel. [1]
- (c) (i) Construct a balanced equation illustrating the reaction between methane and steam. [1]
- (ii) State the type of reaction described in (c)(i). [1]
- (d) The half-equation for the process occurring at the left-hand compartment of the ammonia fuel cell is:
- $$2\text{NH}_3 + 6\text{OH}^- \longrightarrow \text{N}_2 + 6\text{H}_2\text{O} + 6\text{e}^-$$
- (i) Write a balanced half-equation for the process that occurs at the right-hand compartment of the ammonia fuel cell. [1]
- (ii) Construct a balanced equation for the reaction in the ammonia fuel cell. [1]
- (e) One of the intermediate species found in the left-hand compartment of the ammonia fuel cell is hydrazine, N_2H_4 .



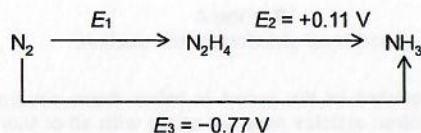
hydrazine



'Eclipsed' Newman projection of hydrazine

- (i) Draw the Newman projection of hydrazine, representing the lowest energy conformation, which allows it to exhibit a net dipole moment. Briefly explain your answer. [2]
- (ii) State and explain how the basicity of N_2H_4 compares to NH_3 . [1]

A Latimer diagram can be used to relate the standard reduction electrode potentials among N_2 , N_2H_4 and NH_3 .



For example, $\text{N}_2 + 6\text{H}_2\text{O} + 6\text{e}^- \ll 2\text{NH}_3 + 6\text{OH}^-$ $E = -0.77 \text{ V}$

(iii) Construct a half-equation for:

- the reduction of nitrogen to hydrazine
- the reduction of hydrazine to ammonia

[1]

(iv) Given that the Gibbs free energy change of each reduction can be related to the standard reduction electrode potential, construct an equation linking E_1 , E_2 and E_3 , and prove that $E_1 = -1.21 \text{ V}$.

[2]

(f) Suggest a reagent to capture the ammonia produced.

[1]

(g) The electrochemical production of ammonia uses ionic liquids as the electrolyte.

(i) Suggest and explain the advantages of using ionic liquids as the electrolyte.

[2]

(ii) State one operational advantage that $\text{P}_{6,6,6,14}$ has over C_4mPyr .

[1]

(iii) C_4mPyrCl can be synthesised from N-methylbutylamine and A. Draw the structure of A and state the type of reaction involved.

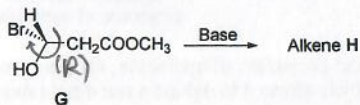
[2]

(h) Draw the molecular orbital diagram of N_2 and fill the orbitals with the appropriate number of electrons. Assume there is no orbital mixing.

[2]

[Total: 22]

2 (a) Compound G undergoes elimination to produce alkene H. This elimination process has a rate law of $\text{rate} = k[\text{G}][\text{Base}]$.



(i) Use Newman projection to draw the conformer of G that will favour the type of elimination mechanism consistent with the rate law.

[1]

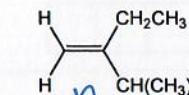
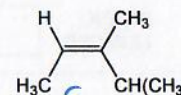
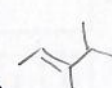
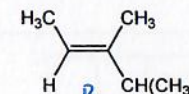
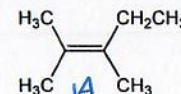
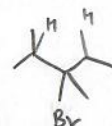
(ii) Outline the mechanism for the formation of alkene H. In your answer, show clearly the stereochemistry of alkene H. You may use B to represent base in your mechanism.

[2]

(iii) Suggest an appropriate base for the above elimination mechanism to occur, in preference over other elimination and substitution mechanism.

[1]

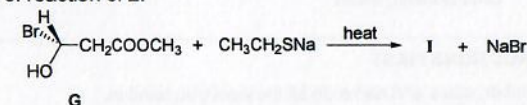
(iv) The following four alkenes are formed from the elimination reaction of 3-bromo-2,3-dimethylpentane.



By considering the type of elimination mechanism that occurred, arrange the four alkenes in increasing order of proportion formed. Explain.

[3]

(b) Compound G also reacts with $\text{CH}_3\text{CH}_2\text{SNa}$ to give compound I. This reaction has an overall order of reaction of 2.



(i) Suggest the type of reaction that took place.

[1]

(ii) Suggest the structure of compound I, showing the stereochemistry.

[1]

(iii) An experiment was conducted to investigate the impact of solvents on substitution reactions. The results were as follows.

solvent	percentage yield of product	time taken
CH_3OH	71 %	20 hours
$(\text{CH}_3)_2\text{SO}$	91 %	20 minutes

Using the results of the experiments, state which is the appropriate solvent for compound G to react with $\text{CH}_3\text{CH}_2\text{SNa}$ such that compound I is the major product.

Using your knowledge of chemical bonding, explain your choice.

[2]

(iv) Comparing $\text{CH}_3\text{CH}_2\text{S}^-$ and $\text{CH}_2\text{C}/\text{CH}_2\text{S}^-$, suggest which nucleophile will provide a higher reactivity with compound G. Explain your answer.

[2]

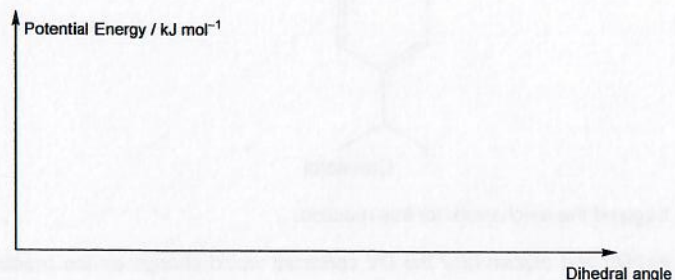
(c) Compound J exhibits conformational isomerism.



On the axes provided below, draw the conformational analysis graph for compound J.

Clearly indicate the stereoisomers required for the respective maxima and minima.

[5]



[Total: 18]

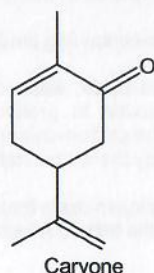
- 3 The specific rotation, $[\alpha]_D$, of a chiral molecule is given by the formula shown.

$$[\alpha]_D = \frac{\alpha}{cl}$$

α is the observed rotation of plane-polarised light
 c is the concentration in g cm^{-3}
 l is the optical path length in dm

The fragrant $\text{C}_{10}\text{H}_{14}\text{O}$ ketone carvone, was isolated from both spearmint and caraway.

All the physical properties of carvone from these two sources seemed to be identical but the odours of the two carvones were different and reflected their source.



- (a) (i) Draw and name the stereoisomers of carvone using R, S convention.

[2]

- (ii) The specific rotation of (S)-carvone is $+61^\circ$ when measured neat (pure liquid sample with no solvent). The optical rotation of a sample of a mixture of R and S carvone is measured at -23° .

Calculate the enantiomeric excess for the sample.

[1]

- (b) R-carvone can be converted to (5R,7R)-8-bromo-7-(2-nitrobenzoate) carvone (compound X) by adding carvone, nitrobenzoic acid and N-bromosuccinimide to a solution of organocatalyst Quinidine (20%). The reaction mixture is stirred for 6 days at room temperature. After that, solvent is evaporated, and the resulting crude product compound X is isolated and purified.

Under alkaline conditions, compound X (M_r : 350.9) forms compound Y (M_r : 166). The following procedure is carried out on a sample of compound X.

- The sample and K_2CO_3 are dissolved in 20 cm^3 methanol at 35°C with stirring.
- The progress of reaction is monitored by thin-layer chromatography (TLC).
- The solvent is evaporated and product mixture containing only X and Y is collected.

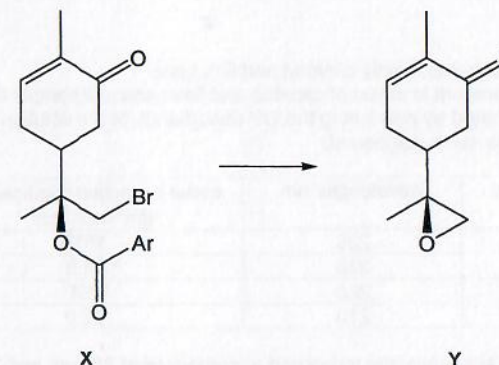


Table 3.1

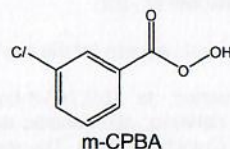
Compound	$[\alpha]_D$
X	$+41.7^\circ$
Y	-13.5°

- 5.97 g of product mixture is added to 10 cm^3 of trichloromethane and a polarimeter cell length of 1.0 dm is filled with the solution.
- The optical rotation is measured.

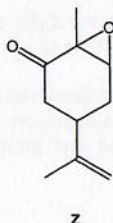
It is found that at equilibrium the observed rotation of the solution is $+10.07^\circ$.

- (i) Calculate the specific rotation of the equilibrium mixture. [1]
- (ii) Calculate the concentration of Y in CHCl_3 solution in g cm^{-3} . [1]
- (iii) Draw a mechanism for the conversion of X to Y. [2]

- (iv) When meta-chloroperoxybenzoic acid (m-CPBA) is added to carvone under suitable condition, compound **Y** is formed.



Suggest why compound **Z** cannot be formed by using m-CPBA as a reagent.



[1]

- (c) Caraway oil consists mainly carvone and limonene. The relative amount in moles of carvone and limonene in a sample of caraway oil may be estimated by measuring the UV absorbance of the solution at a suitable wavelength for each compound.

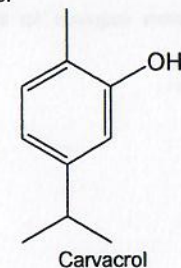
compound	wavelength/ nm	molar extinction coefficient, ϵ / $\text{dm}^3 \text{mol}^{-1} \text{cm}^{-1}$
carvone	235	1513
	318	41.9
limonene	220	251
	210	22.9

For a particular sample, the measured absorbances at 235 nm and 220 nm were 0.89 and 0.11 respectively.

- (i) Explain the principles of UV spectroscopy of organic molecules. [2]
- (ii) Assign the modes of electronic transitions which give rise to the bands in carvone. [1]
- (iii) Calculate the ratio of the relative amounts in moles of the two compounds in the sample. [1]
- (iv) Sketch the spectrum that you would expect for the solution between 200 nm and 300 nm. [1]

Use your sketch to suggest why the ratio calculated in (c)(iii) might be inaccurate. [2]

- (v) Carvone undergoes isomerism in the presence of sulfuric acid to form carvacrol. This process involves the generation of a tertiary carbocation and rearrangement reactions.



Suggest the mechanism for this reaction. [3]

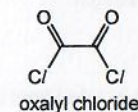
- (vi) Predict and explain how the UV spectrum would change as the reaction proceeds. [1]

[Total: 20]

Section B

Answer two questions from this section

- 4 Hydrogen chloride gas can be produced from the reaction between oxalyl chloride and water.

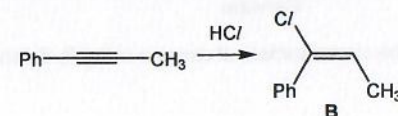


- (a) When oxalyl chloride and water react in the mole ratio of 1:1, two carbon-containing products are also formed in the reaction.

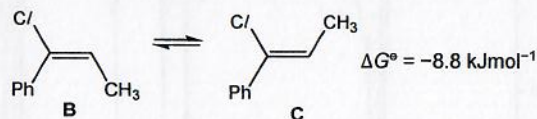
State the identities of the two carbon-containing products. [1]

By mixing oxalyl chloride with aluminium oxide, water adsorbed on the surface of aluminium oxide reacts with oxalyl chloride to produce hydrogen chloride. The hydrogen chloride can be added across the carbon-carbon triple bond to give alkenes. This addition reaction can be used to study the isomerization of alkenes.

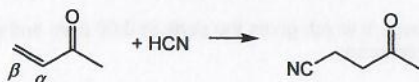
When 1-phenylpropyne is mixed with aluminium oxide that has been treated with oxalyl chloride, the following reaction occurs in the first 10 minutes.



After 10 minutes, the proportion of **B** falls as it isomerises to **C** and the proportion of **C** increases, until an equilibrium is reached between both alkenes.



- (b) (i) Assign the E/Z configuration for **B** and **C**. [1]
- (ii) Explain why only **B** forms initially. [1]
- (iii) Given that $\Delta G^\circ = -RT \ln K_c$, determine:
- the K_c value for the equilibrium between **B** and **C**.
 - the percentage of **B** in the equilibrium mixture. [2]
- (c) **B** reacts with hydrogen chloride to form a carbocation intermediate which subsequently produces **C**.
- (i) Two possible carbocations can be formed in the reaction described in (c).
Draw the carbocation that is more likely to be formed and explain your choice. [2]
- (ii) Using Newman projection, draw the two conformations of the carbocation intermediate that gives to **B** and **C** respectively. You may represent the benzene ring as Ph in your answer. [2]
- (iii) Using your answer to (c)(ii), explain:
- how isomerisation of **B** to **C** occurs,
 - why **C** is formed in larger proportions than **B** in the equilibrium mixture. [3]
- (d) Butenone can react with HCN according to the reaction:



A small amount of potassium hydroxide is needed to start the reaction.

- (i) State the type of reaction illustrated above. [1]
- (ii) Explain why:
- the β – carbon atom is more electron deficient than the α – carbon atom
 - the nucleophile attacks the β – carbon atom and not the carbon atom of the ketone functional group. [3]

- (iii) Propose a mechanism for the reaction in (d). [2]
- (iv) Draw the structures of two organic side products. [2]

[Total: 20]

- 5 (a) Pyrroles are building blocks in pharmaceutical chemistry. Thus, research has been ongoing to synthesise pyrroles in a single step from cheap and readily available materials.

Fig. 5.1 below shows the synthetic route of pyrroles from a modified amino acid, *trans*-4-hydroxyproline, in the presence of acid catalyst and heat.

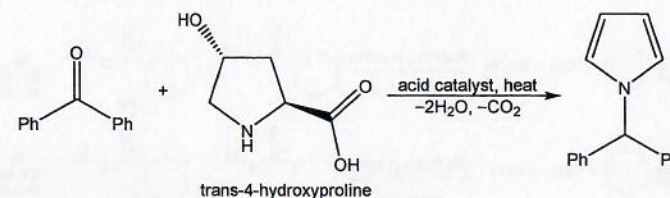
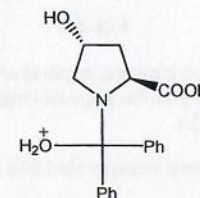


Fig. 5.1

- (i) Suggest a possible mechanism for this reaction given that it forms an intermediate as shown below:



- (ii) The reaction can be followed using IR spectroscopy.

Suggest and explain the major differences between the IR spectra of the starting compound $(\text{Ph})_2\text{CO}$ and the final product shown in Fig. 5.1. [2]

- (b) Active research on carbonylation using CO is still ongoing as it is more reactive than CO_2 .

In recent years, Beller and co-workers developed methods to generate CO from CO_2 in situ (i.e. produced freshly within the reaction flask). The CO generated in situ is immediately made to insert between the alkene and the alcohol.

Fig. 5.2 illustrates one such reaction.

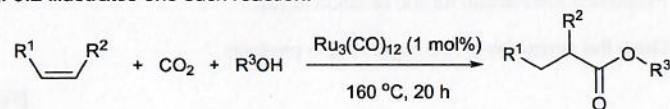


Fig 5.2

To prove that CO was generated in-situ, a series of isotope labelling experiments were carried out.

The experimental results of three experiments are shown in Fig. 5.3.

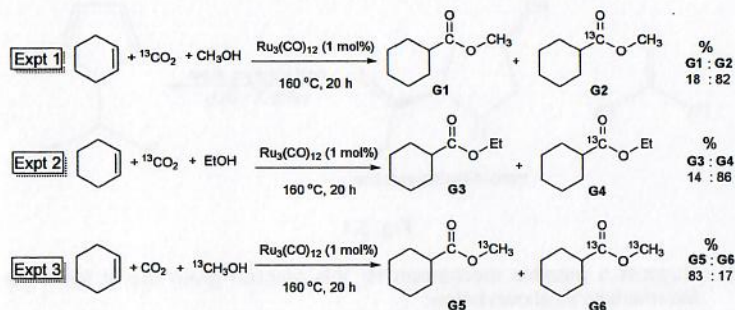
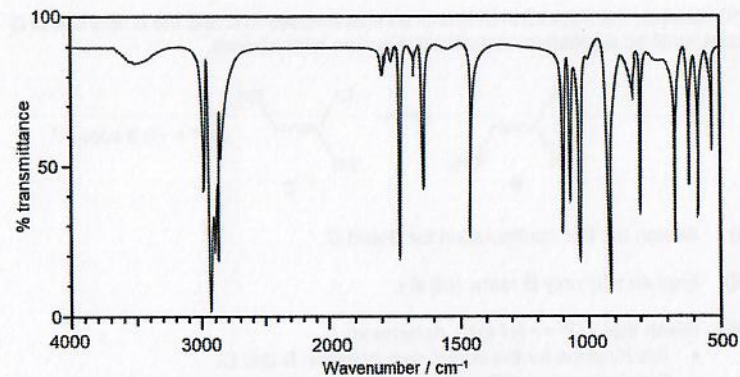


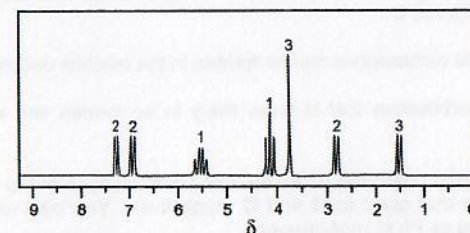
Fig. 5.3

- (i) Suggest how the isotope labelling experiments in Fig. 5.3 indicate that the carbonyl group inserted into the product largely originate from CO_2 and not from the CO in $\text{Ru}_3(\text{CO})_{12}$. [2]
- (ii) Suggest how two ^{13}C were incorporated into G6. [1]
- (c) An unknown compound **U** was isolated from root extract which was shown to have medicinal property and analysed for its structure.

Compound **U** has a molecular ion at m/z value of 218, with relative abundance of 67.1%, and a peak at m/z value of 219 with a relative abundance of 9.5%. It is found that **U** is made of 22% oxygen by mass. The IR spectrum of compound **U** is as shown.



The ^1H NMR spectrum of a sample of **U** dissolved in CDCl_3 is shown below.



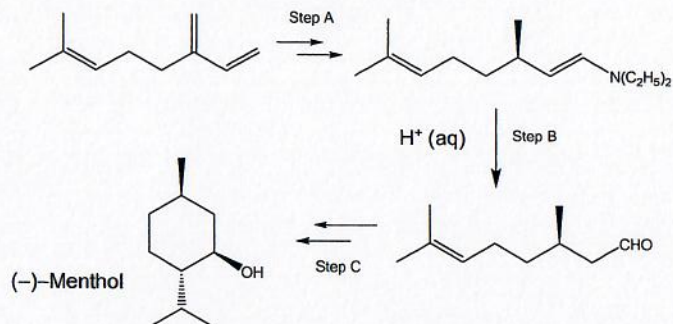
Upon heating compound **U** under reflux with dilute H_2SO_4 , compound **V** was obtained as the only organic product. When aqueous sodium carbonate is added to separate samples of compounds **U** and **V**, effervescence is observed only for compound **V**.

Both compounds **U** and **V** decolourise orange aqueous bromine.

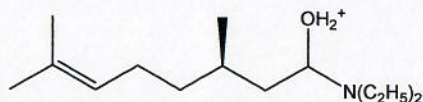
- (i) Identify the reagent which gives the peak at 0.00 ppm and explain its role in ^1H NMR spectroscopy. [2]
- (ii) Use the spectroscopic data and other information provided above, deduce the structure of compound **U**. Explain your answer fully. [9]

[Total: 20]

- 6 (a) (–)-Menthol, an organic molecule, is a major component of the taste and fragrance of peppermint and is widely used in cakes, cosmetics and tooth powder. It is now synthesised industrially as shown below.

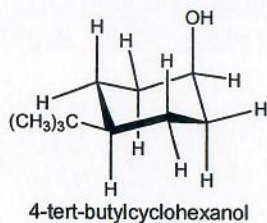


- (i) Draw the mechanism for Step B where an enamine is converted into an aldehyde. One of the intermediates in this mechanism is given below.



- (ii) Describe how you would use IR spectroscopy to verify the complete conversion of intermediate to (–)-Menthol in Step C. [3]
- (iii) A constitutional (structural) isomer of (–)-Menthol is 4-tert-butylcyclohexanol. [2]

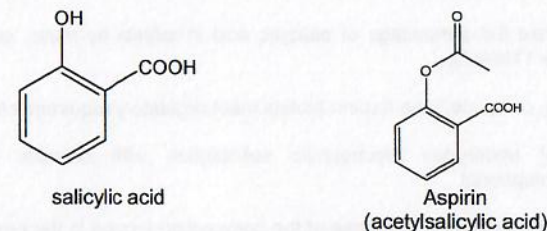
There are a total of 7 signals in ^1H NMR spectrum of 4-tert-butylcyclohexanol in CDCl_3 . Describe and interpret these 7 signals.



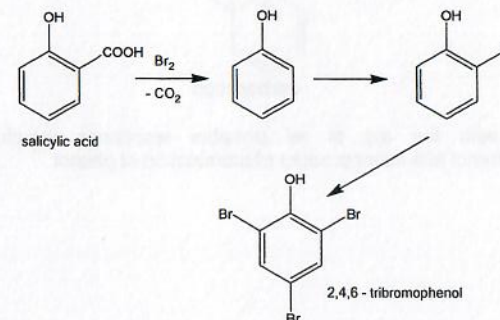
- (iv) Suggest a constitutional (structural) isomer of (–)-Menthol that will give the least number of signals in ^1H NMR spectrum using the same solvent. [1]

- (b) Another useful organic molecule is Aspirin, acetylsalicylic acid. It is used to treat pain and to reduce fever, at low doses, to prevent heart attacks, strokes and blood clot formation.

Aspirin tablets can be contaminated with salicylic acid, the reagent for synthesis of acetylsalicylic acid. The content of salicylic acid is controlled in accordance with national pharmacopoeias. It must not exceed 0.1 – 0.3%.



Redox titration using bromine as an oxidising agent is one of the possible techniques to determine salicylic acid in Aspirin. Aspirin does not react with bromine.



Excess bromine is determined by titration against solution of NaAsO_2 .

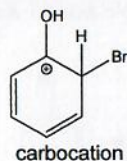
Redox titration:

- A portion of 4.4035 g of powdered Aspirin tablets was dissolved in 250 cm^3 volumetric flask and top up to mark with deionised water.
- Br_2 was generated in situ (in the same glassware) by mixing 0.5950 g of potassium bromide and 0.3340 g of potassium bromate in 100 cm^3 of water in the presence of few drops of concentrated sulfuric acid. This aqueous mixture was labelled as W.
- 25.0 cm^3 aliquot, containing Aspirin, was transferred to a 250 cm^3 dilution flask. 100 cm^3 of W was added.

- The contents in the dilution flask was made up to 250 cm³ with water. The flask was fitted with a stopper and shaken for 10 minutes. The resulting solution was labelled as **Solution Z**.
- Solution Z** was titrated against a solution of 0.02015 mol dm⁻³ NaAsO₂. 25.0 cm³ of **Solution Z** gives a mean titre of 9.93 cm³.

- (i) Write the ionic equation of the reaction of bromide, Br⁻, and bromate, BrO₃⁻, to produce bromine. [1]
- (ii) Calculate the percentage of salicylic acid in tablets by mass, given that 1 Br₂ ≡ 1 NaAsO₂. [3]
- (iii) Hence, conclude if the Aspirin tablets meet regulatory requirements. [1]
- (iv) Phenol undergoes electrophilic substitution with bromine to give 2-bromophenol.

Draw all resonance structures of the carbocation formed in the bromination of phenol. One of which is given below.



- (v) Explain, with the aid of all possible resonance structures, why 3-bromophenol is a minor product of bromination of phenol. [3]

[Total: 20]



RIVER VALLEY HIGH SCHOOL

JC 2 PRELIMINARY EXAMINATION

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H3 CHEMISTRY

9813/03

Paper 1

23 September 2022

INSERT

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your Centre number, class and name on all the work you hand in.

This insert contains information for Question 1. **Do not write your answers on the insert.**

Information for Question 1

Various countries have started moving towards the use of green fuels, in an attempt to reduce carbon emissions into the atmosphere.

Extract 1 compares hydrogen and ammonia as potential green fuels.

Extract 1 (adapted from G. Jeerh et al., *J. Mater. Chem. A*, 2021, 9, 727 – 752)

Hydrogen is used in the hydrogen fuel cell and it produces water, an environmentally friendly product. It is currently extracted mainly via the steam reformation of natural gas which largely comprises methane. However, this method of production is not sustainable as it releases large amounts of carbon dioxide into the atmosphere. The storage of hydrogen through liquefaction has also been a major challenge for many economies to adopt it.

Ammonia's production via the Haber-Bosch process is energetically intense, accounting for 2% of global energy consumption. Part of the reason for this high energy consumption is the need for hydrogen as a raw material. However, its potential for energy storage has been recognised, with the development of various ammonia fuel cells.

Table 1 summarises some properties of hydrogen and ammonia.

Table 1

property	hydrogen (gas)	ammonia (liquid)
density / kg m^{-3}	X	682
energy density / MJ dm^{-3}	2.76	12.9
pressure / MPa	35.0	-
temperature / K	298	298

key: $\text{MPa} = 1 \times 10^6 \text{ Pa}$

Fig. 1.1 shows a simplified ammonia fuel cell.

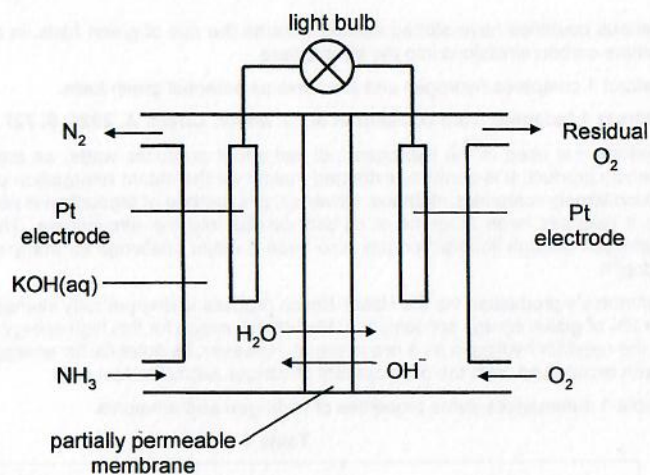


Fig. 1.1

Extract 2 (adapted from F Zhou et al., *Energy Environ. Sci.*, 2017, 10, 2516 – 2520)

Renewable energy can be used to drive the electrochemical production of ammonia. A simplified diagram of this process is shown in Fig. 1.2.

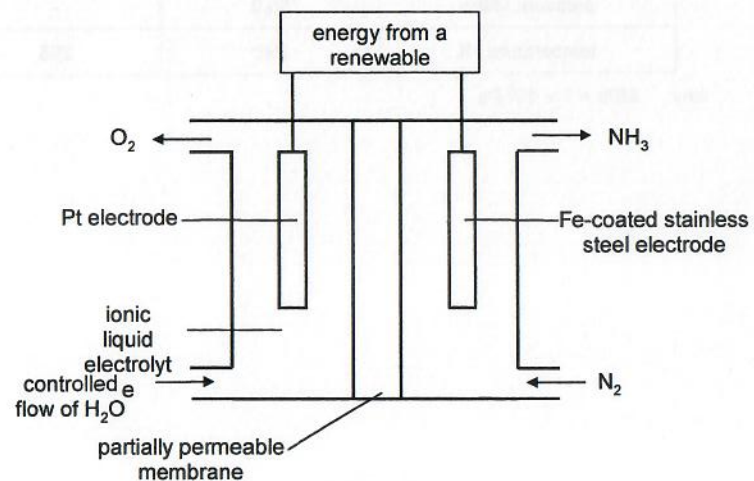


Fig 1.2

- trihexyltetradecylphosphonium-tris(pentafluoroethyl)trifluorophosphate [$\text{P}_{6,6,6,14}$][eFAP],
 - 1-butyl-1-methylpyrrolidinium-tris(pentafluoroethyl)trifluorophosphate [C_4mPyr][eFAP]
- were used and studied as separate electrolytes for this process.

Fig. 1.3 shows the structures of these ions.

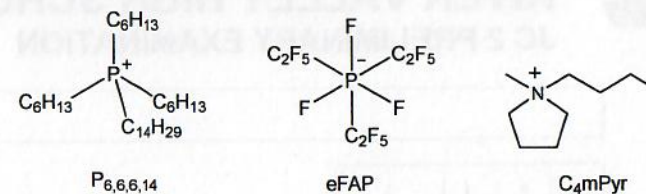


Fig. 1.3

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